# Project Initialization and Planning Phase

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| **Date** | 16 October 2026 |
| **Team ID** | Roll 58 |
| **Project Title** | EV Market project planning phase |
| **Maximum Marks** | 3 Marks |

## Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

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| An interactive Tableau Public dashboard that summarizes electric-vehicle market data to help users explore sales, vehicle categories, battery technologies, range, pricing, customer ratings, and manufacturing geography. | |
| **Objective** | Transform the EV dataset into clear, interactive visual insights so users can compare manufacturers and vehicle segments, identify patterns, and make data-informed observations. |
| **Scope** | Includes data inspection and preparation, KPI cards, eight visualizations, dashboard layout, filters, and a concise insights summary. It is an academic analysis prototype, not a live market-data feed or a forecasting system. |
| EV market data contains multiple attributes and can be difficult to compare when viewed only as raw rows or separate charts. Users may struggle to quickly understand which manufacturers or categories lead in sales, how range and price vary, and how battery types and manufacturing countries are represented. | |
| **Description** | The project uses the provided EV dataset (CSV/XLSX) and a Tableau dashboard to organize important fields such as manufacturer, annual sales units, category, battery type, range, price, customer rating, manufacturing country, and model year. Field names should be checked against the final source file. |
| **Impact** | A well-organized dashboard reduces manual comparison effort, makes patterns easier to communicate, supports classroom/business discussion, and helps users ask more focused questions about the EV market. Conclusions depend on dataset quality and coverage. |
| Inspect the dataset; validate field names and data types; handle missing/duplicate values where appropriate; create calculated measures and aggregations; build individual worksheets; assemble KPI cards and charts into a dashboard; add relevant filters; validate totals and labels; publish through Tableau Public if required. | |
| **Key Features** | KPI cards for key metrics; annual sales by manufacturer; sales by category; battery technology distribution; average range by category; price-versus-range scatter plot; customer ratings by manufacturer; manufacturing by country; sales by model year; interactive filters and readable tooltips. |
| Key Features | Highlight the unique aspects of the proposed solution |

## Resource Requirements

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| Hardware | Computing resources | Personal laptop/desktop; no dedicated GPU required for Tableau dashboard development. |
| Recommended 8 GB RAM or more for smoother spreadsheet/Tableau use; minimum depends on local setup. | | |
| Hardware | Storage | Enough free disk space for the dataset, Tableau workbook, exported images, and documentation; approximately 1–2 GB is generally sufficient for this prototype. |
| Software | Frameworks | Tableau Public / Tableau Desktop (as available); spreadsheet application for data inspection. |
| Software | Libraries | No coding libraries are required for the dashboard itself. Optional Python with pandas/NumPy may be used for data validation or cleaning. |
| Tableau Public, spreadsheet editor, web browser, and a folder for versioned project files; optional Git for version control. | | |
| Data | Dataset | Provided EV market dataset, currently described as 200,000 rows and 37 columns. Confirm exact row/column count and field names in the final submitted source file. |
| Libraries | Additional libraries | e.g., scikit-learn, pandas, numpy |
| Development Environment | IDE, version control | e.g., Jupyter Notebook, Git |
| **Data** | | |
| Data | Source, size, format | e.g., Kaggle dataset, 10,000 images |